

Dummy RAPM: Representing Low-Minute Players in Regularized Adjusted Plus-Minus

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Abstract

Regularized Adjusted Plus-Minus (RAPM) estimates player associations with scoring margin from stint-level lineup indicators. When low-minute player columns are removed while their stints remain in the sample, the reduced design no longer records the complete lineup. We introduce Dummy RAPM, which augments the retained-player matrix with five low-minute-player count indicators for each lineup side. The ten indicators receive a common ridge penalty distinct from the player penalty. We evaluate the model over 17 NBA seasons using chronological inner validation and an untouched March–April test period. Inner validation selects a 10-minute-per-appearance threshold and a dummy-to-player penalty ratio of 2.2. Dummy RAPM lowers mean game-margin RMSE from 12.884 to 12.843, an average gain of 0.041 points (0.30%), and improves RMSE in 14 of 17 seasons. The gain is modest in magnitude but consistent across paired season-level tests. The results support a partially pooled representation of lineup-side counts, while providing little evidence that filtering players without representing them improves prediction.

1 Introduction

Basic plus-minus measures a team’s score differential while a player is on the court but does not adjust for teammates or opponents. Adjusted Plus-Minus (APM), associated with the WINVAL work of Jeff Sagarin and Wayne Winston and described publicly by [Rosenbaum \(2004\)](#), addresses this limitation by regressing stint-level score differentials on player indicators.

APM estimates are unstable when players repeatedly share the floor with the same teammates and opponents or appear in few stints. Ridge regression stabilizes correlated designs by shrinking coefficients ([Hoerl and Kennard, 1970](#)). [Sill \(2010\)](#) applied this approach to APM and emphasized evaluation on future games, yielding Regularized Adjusted Plus-Minus (RAPM). Bayesian partial-pooling approaches provide a related response to weak identification. [Fearnhead and Taylor \(2011\)](#) estimated offensive and defensive abilities while borrowing information across seasons, and [Deshpande and Jensen \(2016\)](#) modeled changes in win probability using Bayesian linear regression. Box Plus-Minus (BPM) is a separate box-score-based branch rather than a Bayesian plus-minus estimator itself. However, the BPM 2.0 regression was trained against Bayesian-prior-informed, multiyear RAPM targets ([Myers, 2020](#)).

Some RAPM implementations remove low-minute players before fitting. Removing a column does not remove the corresponding stints from the response, so the omitted contribution remains in the response. It can be attributed to retained-player coefficients when it is correlated with

retained-player indicators; otherwise it is absorbed primarily by the intercept and residual. We ask whether a small set of partially pooled count indicators can represent this missing lineup information without estimating an individual coefficient for every low-minute player.

Our contribution is methodological and empirical. We construct separate count indicators for excluded home and away players, assign them a ridge penalty distinct from the player penalty, and evaluate the resulting model using chronological nested selection. We also impose explicit lineup, scoring, and game-margin reconstruction checks before fitting.

2 Model

2.1 Stint response and player design

A stint is an interval over which neither lineup changes. Let Δ_i be the change in home-minus-away score margin during stint i , and define exposure

$$q_i = \frac{P_{H,i} + P_{A,i}}{2},$$

where $P_{H,i}$ and $P_{A,i}$ count home and away possession-ending events. The response is

$$y_i = 100 \frac{\Delta_i}{q_i},$$

the home-team margin per 100 team possessions. Observation i receives weight q_i , so a longer stint contributes more to estimation. Scaling before estimation makes coefficients directly interpretable in points per 100 team possessions.

For retained player j , $x_{ij} = 1$ when the player is in the home lineup, $x_{ij} = -1$ when in the away lineup, and $x_{ij} = 0$ otherwise. The ordinary no-dummy RAPM estimator is proportional to

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \left\{ \sum_{i=1}^n q_i (y_i - \alpha - \mathbf{x}_i^\top \beta)^2 + \lambda_P \|\beta\|_2^2 \right\}.$$

The intercept is not penalized, and predictors are not standardized. Models are fit with `glmnet` (Friedman et al., 2010).

2.2 Lineup-side count indicators

For $k = 1, \dots, 5$, define

$$d_{H,i}^{(k)} = \mathbb{1}\{\text{the home lineup in stint } i \text{ contains exactly } k \text{ excluded players}\},$$

$$d_{A,i}^{(k)} = \mathbb{1}\{\text{the away lineup in stint } i \text{ contains exactly } k \text{ excluded players}\}.$$

Count zero is the reference category. The augmented model therefore contains ten dummy regressors, five for each lineup side. Separate player, home-dummy, and away-dummy namespaces prevent numeric player identifiers from colliding with dummy names. A five-column signed encoding would impose equal-and-opposite home and away coefficients at each count. The ten-column encoding does not impose that restriction: a given number of excluded home players and excluded away players can have different magnitudes rather than being forced to differ only in sign.

Let γ_H and γ_A denote the two five-element dummy blocks. Dummy RAPM estimates

$$\arg \min_{\alpha, \beta, \gamma_H, \gamma_A} \sum_{i=1}^n q_i \left(y_i - \alpha - \mathbf{x}_i^\top \beta - \mathbf{d}_{H,i}^\top \gamma_H - \mathbf{d}_{A,i}^\top \gamma_A \right)^2 + \lambda_P \|\beta\|_2^2 + \lambda_D (\|\gamma_H\|_2^2 + \|\gamma_A\|_2^2).$$

We tune the relative penalty $\rho = \lambda_D / \lambda_P$. A common dummy-specific penalty permits a different degree of pooling from player coefficients without introducing ten sparsely estimated hyperparameters.

2.3 Low-minute threshold

Rosenbaum (2004) placed players with less than 250 minutes in the two seasons combined into a common reference category. Our threshold instead uses mean minutes conditional on appearing in a game, calculated only from the relevant training period. This avoids labeling an ordinary rotation player as low-minute solely because of missed games, but it differs from a total-minutes rule and can retain players with few appearances. We therefore interpret the threshold as a prediction-tuned modeling choice rather than a definition of player quality.

3 Data Construction and Validation

We obtain publicly accessible ESPN play-by-play and player box-score records through the open-source `hoopR` package (Gilani, 2026). Official box-score starter flags initialize each five-player lineup; recorded substitutions update the lineup thereafter. If either team lacks exactly five unique recorded starters, we exclude the full game rather than guess an opening lineup. Across the primary sample, 34 games are excluded for this reason, including 19 in 2019.

Every modeled stint must contain exactly five unique home players and five unique away players. Stint points are constructed from changes in the reported home-minus-away scoreboard margin, including the first scoring event in each stint. This scoreboard-delta construction credits technical free throws and other free-throw points even when an event is not marked as ending a possession. A scoring segment with zero possession exposure—for example, free throws separated by a substitution—is joined to the next positive-exposure stint, or to the preceding one at the end of a game. Thus no scored point is assigned an infinite per-possession response. The reconstructed stint margins must sum exactly to the official final margin in every included game. Period labels are also required to form a contiguous sequence beginning at one; this removes a single disconnected postgame record labeled as period 29 without truncating valid multi-overtime games.

The primary analysis retains all possessions, including those labeled as garbage time and those played in overtime. Removing them would change the full-game prediction target. The analysis uses open-source software, but we describe the underlying ESPN data as publicly accessible rather than fully open source.

Table 1 reports the season-level sample. The analysis contains 20578 games and 519,745 positive-exposure stints. “Out” denotes regular-season games in May–September, which are excluded before model fitting or evaluation. Starter exclusions may overlap the out-of-window count.

4 Selection and Evaluation

The sample contains 17 seasons: 2007–2019 and 2022–2025. It begins in 2007 because starter metadata are incomplete in earlier seasons. The disrupted 2019–20 season and the COVID-shortened 2020–21 season are excluded because neither supports the same October–April chronological split.

Table 1: Season-level analysis sample and chronological windows

Season	All	Oct–Dec	Jan–Feb	Oct–Feb	Mar–Apr	Out	Starter excl.	Stints	Players
2007	1214	443	404	847	362	0	5	28,257	454
2008	1217	448	411	859	358	0	0	27,243	451
2009	1228	471	409	880	348	0	0	26,348	445
2010	1225	470	409	879	346	0	0	26,835	441
2011	1226	479	408	887	339	0	0	27,583	452
2012	990	56	466	522	467	0	1	23,170	478
2013	1226	453	402	855	366	0	5	23,607	469
2014	1225	464	403	867	357	0	1	23,700	482
2015	1226	477	393	870	356	0	0	25,134	492
2016	1227	489	399	888	339	0	0	25,155	477
2017	1230	506	387	893	336	0	1	25,032	484
2018	1226	540	375	915	309	0	2	34,393	539
2019	1230	532	376	908	303	0	19	34,502	537
2022	1230	527	394	921	309	0	0	42,133	605
2023	1230	546	385	931	299	0	0	42,531	539
2024	1231	481	405	886	345	0	0	41,552	572
2025	1231	485	403	888	343	0	0	42,570	569

Within each season, October–December games form the inner training period, January–February games form the inner validation period, and March–April games form the outer test period. Outer training is exactly the union of the October–December and January–February windows. May–September games are in none of the four analysis sets. Programmatic checks require every game to receive one disjoint base-window label and require the latest training date to precede the earliest validation or test date.

The initial inner search considers average-minutes-per-appearance thresholds

$$\{0, 5, 10, 15, 20, 25, 30\}$$

and dummy-to-player penalty ratios

$$\{0, 0.1, 0.25, 0.5, 1, 2\}.$$

At threshold zero, all players observed in the relevant training period are retained. The augmented fit is implemented by reusing the no-dummy fit and appending ten exactly zero dummy coefficients. Its predictions, selected lambda, RMSE, and R^2 are therefore required to equal the no-dummy quantities exactly. It remains a training-observed baseline rather than a claim that every possible player is known in advance.

The broad grid selects a 10-minute threshold. We then evaluate the following penalty grid at that threshold:

$$\rho \in \{1.0, 1.1, \dots, 2.5\}.$$

If a boundary value wins, the grid is extended by increments of 0.1 until the optimum is interior. The checked grid runs from 1 to 2.5 and selects $\rho = 2.2$. All selection uses mean inner-validation game-margin RMSE across seasons. To avoid selecting a threshold for only one member of the final comparison, the threshold score averages Dummy RAPM and no-dummy RAPM RMSE. Exact ties

prefer the lower minute threshold and then the larger dummy penalty. Neither outer RMSE nor R^2 participates in selection.

After selection, both models are refit using all games through February and evaluated once on March–April games. Player eligibility is recomputed using only the corresponding training period. One deterministic five-fold, game-grouped assignment is constructed for each season and training window, using a seed based only on season and window. The same assignment is passed to every threshold, model, and penalty ratio. Within each fit, λ_P is chosen by minimizing cross-validated aggregated game-margin RMSE. Grouping prevents stints from the same game from entering both sides of a fold; common folds ensure that candidate specifications face the same validation split.

For game g , stint predictions are converted back to points and summed:

$$\hat{\Delta}_g = \sum_{i \in g} \hat{y}_i q_i / 100.$$

RMSE is the selection and headline metric. We also report descriptive out-of-sample

$$R_{\text{oos}}^2 = 1 - \frac{\sum_{g \in \mathcal{T}} (\Delta_g - \hat{\Delta}_g)^2}{\sum_{g \in \mathcal{T}} (\Delta_g - \bar{\Delta}_{\mathcal{T}})^2},$$

where $\mathcal{T}$ is the March–April test set and the comparator predicts that season’s mean test-set margin for every game. Paired season-level RMSE differences are summarized with two-sided paired t , Wilcoxon signed-rank, and sign tests. These are descriptive comparisons across seasons, not independent experimental replications.

The exercise conditions on the realized held-out lineups. It evaluates out-of-sample lineup prediction and should not be interpreted as a pregame forecast.

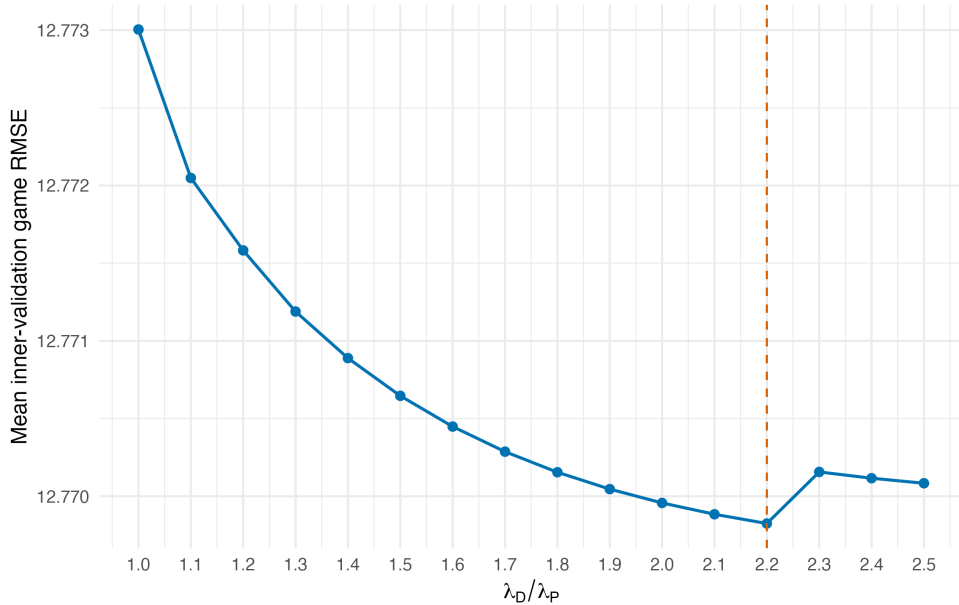


Figure 1: Mean inner-validation game-margin RMSE over the refined dummy-to-player penalty grid. The dashed line marks the selected ratio, 2.2.

5 Results

5.1 Out-of-sample prediction

Table 2: Outer-test predictive performance across 17 seasons

Model	Mean RMSE	Mean R^2	RMSE wins
Dummy RAPM	12.843	0.174	14
No-dummy RAPM	12.884	0.169	3

Dummy RAPM reduces mean game-margin RMSE by 0.041 points, or 0.30%, with a season-level 95% interval from 0.015 to 0.068. The two-sided paired t -test gives $t(16) = 3.310$ and $p = 0.0044$. The two-sided Wilcoxon signed-rank test gives $W = 136.0$ and $p = 0.0052$; the two-sided sign test observes 14 positive differences in 17 seasons and gives $p = 0.0127$. As a secondary descriptive result, mean out-of-sample R^2 changes by 0.005. The RMSE difference should be read together with its small absolute magnitude rather than as evidence of a large practical effect.

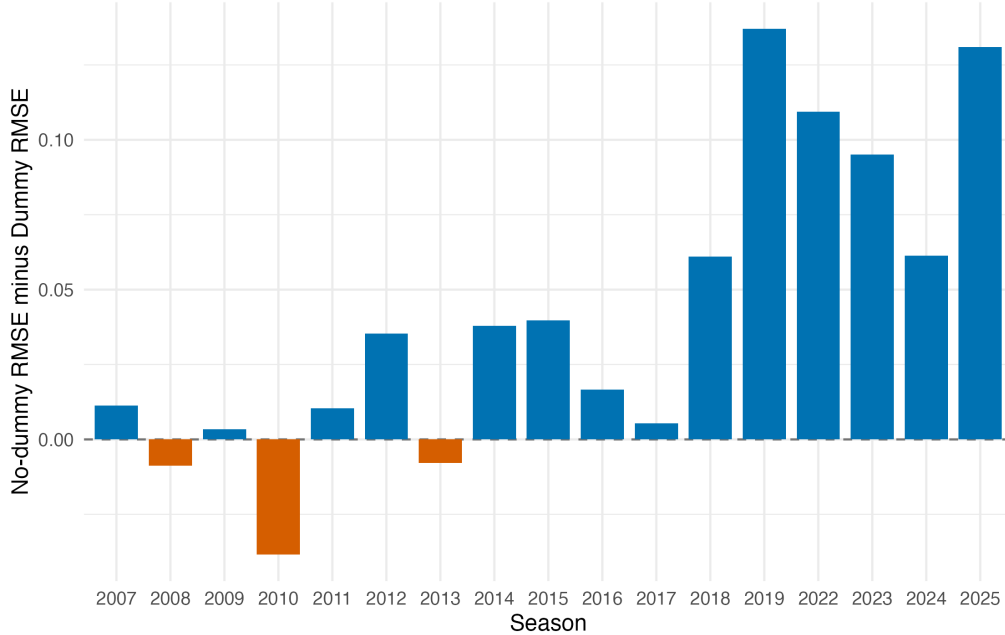


Figure 2: Outer-test game-margin RMSE difference by season. Positive values indicate lower RMSE for Dummy RAPM.

Dummy RAPM has lower RMSE in 10 of 13 seasons through 2019 and in 4 of 4 seasons from 2022 onward. Because 2019 has the largest starter-data exclusion count, we also remove that season as a sensitivity check. The remaining 16 seasons retain an average RMSE gain of 0.035 points ($p = 0.0084$).

5.2 Filtering and representation

The result does not imply that filtering alone improves RAPM. Relative to the zero-threshold no-dummy baseline, the selected no-dummy model is worse by 0.014 RMSE on average ($p = 0.1125$).

Dummy RAPM improves on that baseline by 0.027 RMSE, but this broader comparison is less precise ($p = 0.0445$). The stronger finding is the controlled comparison at the same selected threshold: representing excluded-player counts performs better than simply dropping those columns.

5.3 Dummy coefficients

Table 3: Selected outer-training dummy-category exposure across seasons

Side	Count	Stints	Exposure	Seasons	Seasons present
Home	0	353,707	1,303,375.0	17	2007–2019, 2022–2025
Home	1	20,810	63,360.5	17	2007–2019, 2022–2025
Home	2	2,533	8,203.5	17	2007–2019, 2022–2025
Home	3	691	2,857.5	13	2007, 2012–2019, 2022–2025
Home	4	259	1,101.0	7	2012, 2018, 2019, 2022–2025
Home	5	45	202.0	4	2022–2025
Away	0	351,609	1,297,976.0	17	2007–2019, 2022–2025
Away	1	22,418	67,852.0	17	2007–2019, 2022–2025
Away	2	2,935	9,255.5	17	2007–2019, 2022–2025
Away	3	769	2,738.5	14	2007, 2010, 2012–2019, 2022–2025
Away	4	269	1,096.5	8	2012, 2016, 2018, 2019, 2022–2025
Away	5	45	181.0	4	2022–2025

Table 3 reports how often each home and away count occurs in the selected outer-training designs. Exposure is the sum of $q_i = (P_{H,i} + P_{A,i})/2$ over contributing stints. The season lists distinguish structural zeros caused by an absent category from coefficients merely shrunk close to zero.

Table 4: Selected dummy coefficients across seasons

Lineup side	Low-minute players	Mean coefficient	Directional seasons
Home	1	-1.51	17 negative
Home	2	-0.61	17 negative
Home	3	-0.36	13 negative, 4 zero
Home	4	-0.07	5 negative, 2 positive, 10 zero
Home	5	-0.04	4 negative, 13 zero
Away	1	0.78	3 negative, 14 positive
Away	2	0.24	7 negative, 10 positive
Away	3	-0.05	12 negative, 2 positive, 3 zero
Away	4	-0.07	5 negative, 3 positive, 9 zero
Away	5	0.02	1 negative, 3 positive, 13 zero

The dummy coefficients summarize regularized lineup-level associations and do not estimate causal effects for excluded players. Low-minute lineups may be selected in response to score state, injuries, matchups, or schedule conditions. Relative to zero excluded players on the same side, one excluded home player is associated with -1.51 home-margin points per 100 possessions, while one excluded away player is associated with 0.78 points. The home one-player coefficient is negative in 17 seasons; the away coefficient is positive in 14.

The coefficients are not monotone in the count. Counts of four and five are exactly zero in many seasons, indicating rare categories that are strongly pooled toward zero. We therefore do not interpret the dummy sequence as a dose-response curve, and the cross-season intervals in Figure 3 are descriptive summaries of penalized estimates, not classical coefficient confidence intervals.

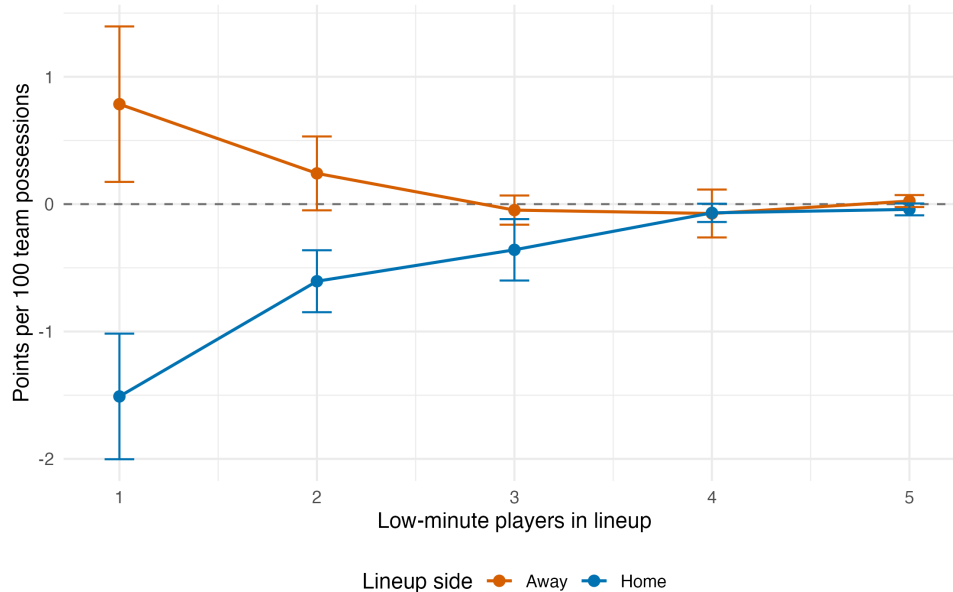


Figure 3: Mean selected home and away dummy coefficients with descriptive 95% cross-season t intervals. Away coefficients retain the home-margin orientation.

6 Discussion

The analysis supports a narrow conclusion: after selecting a 10-minute threshold without using the outer test period, adding ten lineup-side count indicators lowers mean held-out RMSE relative to dropping the same player columns without representation. The focused inner search favors stronger shrinkage for dummy coefficients than for player coefficients. The exact ordering is shallow across the focused grid, and an arbitrarily large dummy penalty would converge toward the no-dummy model.

Several limitations matter. First, the result concerns game-margin prediction, not causal player effects. Second, lineups are reconstructed from event data and starter records rather than an official shift chart. The lineup and score-reconciliation checks catch many errors, but a valid-looking lineup can still be incorrect. Third, the exact penalty optimum is shallow; $\rho = 2.2$ is a reproducible grid choice rather than a sharply identified scientific constant. Fourth, starter metadata require game exclusions concentrated in 2019, and high dummy counts are rare. Finally, adjacent season-level comparisons share players, coaches, and league conditions. The paired tests do not remove this temporal dependence, and future-season validation remains valuable.

7 Conclusion

Dummy RAPM offers a compact way to represent low-minute players whose individual columns are omitted from a RAPM design. In the selected specification, five home and five away count indicators receive a common penalty 2.2 times the player penalty. Across 17 seasons, the model

improves game-margin RMSE in 14 seasons. The average gain is modest, and R^2 is retained only as a secondary description. The evidence favors partially pooling omitted-player counts, not filtering players by itself.

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